# Lab 1 — Finding Similar Things

**Friday, October 16 (~20 min)**

Last updated: October 12, 2026.

In class Friday, October 16 (same day as Lecture 23 / Quiz 6). Quiz 6 is Week 7 matrices — **not** this lab.

**Credit:** Lab points are for effort in class, not for a Canvas file upload. Work in Octave or Python on a laptop. Preview the one-pager; write on the hard copy handed out in class.

Pick **Octave or Python** (same choice as Lab 0).

## Learning objectives

1. Print vectors and report dimensions/length.
2. Compute distances between feature vectors.
3. Name a nearest neighbor and interpret it in context.

## Math connections

Norm and distance (Lecture 07); nearest neighbors (Lecture 09).

## Key syntax (this lab)

| Task | Octave | Python |
|---|---|---|
| Length / shape | `length(x)`, `size(A)` | `len(x)`, `x.shape` |
| Print | `x` or `disp(x)` | `print(x)` |
| Subtract | `x-y` | `x-y` |
| Euclidean norm | `norm(x)` | `np.linalg.norm(x)` |

**Resources:** Lachniet Ch. 1 · Syntax · Resources.

## Ops strip (2–3 min)

Three song feature vectors (loudness, tempo) — units already scaled for today:

$$a = (1, 2), \quad b = (2, 1), \quad c = (4, 3).$$

**C1.0.** Enter `a`, `b`, `c`. Print each and the length of `a`.

**Expected:** length 2.

## Application (~15 min)

**C1.1.** Compute $\|a - b\|$ and $\|a - c\|$ (Euclidean).

**Expected:** $\|a - b\| = \sqrt{2} \approx 1.414$; $\|a - c\| = \sqrt{18} = 3\sqrt{2} \approx 4.243$.

**Octave:** `norm(a-b)`
**Python:** `np.linalg.norm(a-b)`

**C1.2.** Which of $b$ or $c$ is the nearest neighbor of query $a$? One sentence: what does that mean if these were song features?

**Expected:** $b$ is nearer; among these two songs, $b$ is more similar to $a$ in (loudness, tempo).

**C1.3 (scale warning).** Replace $c$ by $c_{\text{bad}} = (4, 30)$ (tempo mistyped $\times 10$). Recompute $\|a - c_{\text{bad}}\|$. Does the nearest neighbor of $a$ among $\{b, c_{\text{bad}}\}$ change?

**Expected:** distance becomes large ($\approx 28.16$); NN is still $b$, but the gap is dominated by the bad scale — feature units matter.

## One-pager

Preview the Lab 1 ticket; write on the hard copy in class.